

Logistics Compliance Intelligence Platform - PRD & Architecture v2

Readable product blueprint updated from compliance automation learnings without shorthand clone positioning

Updated 18 August 2026 | Product category: Logistics Assurance Automation

1. Product positioning

The product is a transport-native continuous assurance platform. It borrows proven product mechanics from modern compliance automation - framework mapping, automated evidence, continuous monitoring, remediation, audit workspaces and governed trust sharing - but applies them to physical logistics operations and multi-party supply-chain assurance.

Positioning guardrail

Do not describe the product using direct shorthand clone analogies. The category we want to create is Logistics Assurance Automation: continuous assurance and audit readiness for logistics operators and outsourced transport supply chains.

2. What we adopt from compliance automation

- Framework-to-control mapping with effective dates and source references.
- Automated evidence collection from connected systems rather than screenshots and spreadsheets.
- Continuous monitoring so control status reflects current operations, not only an annual audit file.
- Findings and remediation workflows with owners, due dates, escalation and closure evidence.
- Audit workspaces that package evidence and support human judgement.
- Trust-sharing mechanics so approved assurance status can be shared with customers and partners under governance.
- Shared-control architecture so one evidence item can support multiple obligations and assurance requests.

3. Why logistics needs a distinct architecture

Area	Generic compliance automation	Logistics assurance automation
Evidence	Policies, cloud configuration, access controls, HR records	Trips, work/rest, telematics, load plans, weighbridge, maintenance, incidents, supplier records

Risk	Digital/security controls mostly within one organisation	Physical transport safety risk across operators, customers, schedulers, sites, carriers and subcontractors
Control effectiveness	Often checks whether controls are configured and evidence exists	Must assess whether transport controls are Present, Suitable, Operating and Effective
Trust sharing	Customer-facing security/trust portal	Compliance Passport for logistics providers with controlled supplier/customer assurance views
Differentiation	Framework coverage and integrations	CoR/HVA applicability, operational data reasoning, evidence lineage and supplier assurance network

4. Core product model: Compliance Control Graph

Every control and test should be represented as structured data so the same graph can power dashboards, findings, remediation, audit packs and supplier passports.

Graph object	Role in product
Regulation / source	Effective-dated source record and jurisdiction.
Obligation	Plain-language duty mapped to role and applicability.
Risk	Unsafe or non-compliant outcome.
Control	Control objective, owner, frequency and status.
Evidence	Data or document proving the control exists, operates or is effective.
Data source	TMS, EWD, GPS, maintenance, HR, WMS, supplier portal or document repository.
Test	Deterministic, analytical or AI-assisted evaluation of evidence.
Finding	Reviewable potential or confirmed exception with severity and PSOE impact.
Remediation	Corrective action, owner, due date, closure

evidence and verification.

Assurance Passport

Customer-facing governed view of selected assurance status and evidence summaries.

5. Applicability engine

The platform must first decide what applies before it scores or tests. This is essential because HVA accreditation is voluntary, while CoR duties can be broader and role-based.

- Jurisdiction and HVNL applicability, with WA/NT handled as future separate rule packs.
- CoR role and level of influence/control over transport activity.
- Vehicle type, activity, route and operating model.
- Accreditation status and HVA module relevance.
- Employee versus subcontracted drivers and supplier chain structure.
- Customer, contract and site-specific obligations.

6. MVP scope

MVP objective

Prove that operational logistics data can automatically create credible compliance evidence, findings, corrective actions and audit-ready workpapers.

- 8 control domains: fatigue, fitness to drive, scheduling/speed, maintenance, mass/dimension, load restraint, driver competency, incidents/CARs.
- 48 structured control tests, maintained in the separate JSON catalogue.
- P0 integrations: TMS, EWD/work diary, telematics/GPS and fleet maintenance; CSV/Excel fallback.
- Core workflows: connect evidence, run tests, raise findings, review findings, create CARs, verify closure and generate audit pack.
- Non-goals: TMS replacement, legal certification, automated auditor judgement or all-regulations coverage.

7. Control-test catalogue summary

Domain	Tests	Example tests	Main evidence	Product phase
Fatigue / Work-Rest	8	Work-time threshold exception; Rest-time deficiency; Missing EWD/work diary record	Driver app, EWD, HR/LMS, TMS	P1
Fitness to Drive	4	Unfit declaration	Driver app, HR,	P1

		followed by dispatch; No pre- start declaration; Post-incident fitness review missing	Incident system, TMS	
Scheduling & Speed Risk	6	Infeasible delivery slot; Schedule pressure plus speeding; Schedule pressure plus fatigue	Customer portal, EWD, TMS, Telematics, WMS	P1
Vehicle Maintenance & Roadworthiness	6	Critical service overdue; Open safety defect at dispatch; Inspection missing before trip	DMS, Driver app, Fleet maintenance, TMS	P1
Mass & Dimension	6	Gross mass exception; Axle group mass exception; Missing weighbridge evidence	Permit register, TMS, Vehicle master, WMS, Weighbridge	P1
Load Restraint	6	Load restraint evidence missing; Photo quality insufficient; Incorrect restraint method flag	Driver app, Incident system, TMS, WMS	P1
Driver Competency	5	Licence expired; Incorrect licence class; Induction missing	Driver app, HR, LMS, Supplier portal, TMS	P1
Incidents & Corrective Actions	6	Incident not classified; CAR overdue; High-risk incident without investigation	Email/DMS, GRC, Incident system, Maintenance, TMS	P1
Supplier / CoR	7	Supplier	Assurance	P2

Assurance

onboarding
evidence missing;
Insurance/certifica
te expired;
HVA/accreditation
status mismatch
platform, Contract
system, DMS,
Procurement/ERP,
Supplier portal

8. Sample detailed control-test records

The full 48-test detail is in the companion JSON catalogue. The records below show the level of detail expected for implementation.

FAT-001 - Work-time threshold exception

Field	Value
Domain	Fatigue / Work-Rest
Obligation	Manage fatigue risk and work/rest compliance for fatigue-regulated heavy vehicle activity.
Risk	Driver operates while fatigued or outside configured work/rest controls.
Control	Fatigue monitoring and intervention control
Evidence	EWD/work diary, roster, trip schedule, training, override record
Data sources	EWD, TMS, HR/LMS, Driver app
Required data fields	driver_id, trip_id, work_start, work_end, rest_period, schedule_start, schedule_end, rule_pack_id, training_status
Test type	Deterministic
Test logic	Calculate actual work time against configured effective-dated fatigue rule pack.
PSOE	Operating/Effective
Severity	High
Finding	Potential fatigue control failure
Remediation	Investigate driver/schedule; adjust roster or trip; capture manager sign-off; monitor recurrence

FIT-001 - Unfit declaration followed by dispatch

Field	Value
Domain	Fitness to Drive
Obligation	Do not allow or encourage drivers to operate while unfit to drive.
Risk	Driver operates while impaired, fatigued, ill or otherwise unfit.
Control	Pre-start fitness declaration and escalation control
Evidence	Driver declaration, dispatch record, incident records
Data sources	Driver app, TMS, Incident system, HR
Required data fields	driver_id, declaration_status, declaration_time, dispatch_time, trip_id, incident_id
Test type	Deterministic
Test logic	Driver declares unfit but receives or starts an assignment.
PSOE	Operating/Effective
Severity	Critical
Finding	Potential fitness-to-drive control failure
Remediation	Stop assignment where needed; supervisor review; update driver status; document corrective action

SCH-001 - Infeasible delivery slot

Field	Value
Domain	Scheduling & Speed Risk
Obligation	Transport arrangements should not create unreasonable safety risk.
Risk	Schedules or customer expectations encourage fatigue, speed or unsafe operations.
Control	Journey feasibility and schedule-risk control
Evidence	Booking, pickup, delivery slot, actual route, load delay, speed/fatigue events

Data sources	TMS, Telematics, EWD, WMS, Customer portal
Required data fields	trip_id, customer_id, site_id, planned_pickup, actual_pickup, delivery_slot, distance_km, gps_events, speed_events, ewd_remaining_hours
Test type	Analytical
Test logic	Available time is below configured feasible journey window.
PSOE	Suitable/Effective
Severity	High
Finding	Potential schedule-induced transport safety risk
Remediation	Re-plan trip; adjust delivery slot; document customer/scheduler decision; monitor recurrence

MNT-002 - Open safety defect at dispatch

Field	Value
Domain	Vehicle Maintenance & Roadworthiness
Obligation	Operate vehicles that meet applicable safety and roadworthiness standards.
Risk	Unsafe or unroadworthy vehicle is dispatched.
Control	Maintenance, defect and release-to-service control
Evidence	Service records, defects, inspections, repair orders, vehicle master
Data sources	Fleet maintenance, Driver app, TMS, DMS
Required data fields	vehicle_id, service_due_date, service_status, defect_id, defect_severity, repair_order, inspection_status, dispatch_time
Test type	Deterministic
Test logic	Vehicle assigned while critical defect remains open.
PSOE	Operating/Effective
Severity	Critical
Finding	Potential vehicle safety control failure

Remediation

Ground vehicle where required; complete repair/inspection; capture release approval; monitor repeat pattern

MDL-001 - Gross mass exception

Field	Value
Domain	Mass & Dimension
Obligation	Meet mass, dimension and loading requirements applicable to the vehicle, load and route.
Risk	Overloaded or incorrectly configured vehicle operates.
Control	Mass verification and permit-condition control
Evidence	Weighbridge record, load plan, permit, vehicle configuration, trip record
Data sources	TMS, WMS, Weighbridge, Permit register, Vehicle master
Required data fields	vehicle_id, trip_id, load_id, gross_mass, axle_mass, permit_id, route_id, vehicle_config, load_plan_id
Test type	Deterministic
Test logic	Calculated/actual gross mass exceeds configured limit/permit.
PSOE	Operating/Effective
Severity	High
Finding	Potential mass/dimension/loading control failure
Remediation	Hold/reconfigure/reweigh load; confirm permit conditions; document approval; monitor site trend

LDR-003 - Incorrect restraint method flag

Field	Value
Domain	Load Restraint
Obligation	Loads must be placed and restrained so the vehicle is not unsafe or unstable and load cannot fall or be

	dislodged.
Risk	Load shifts, falls, destabilises vehicle or creates road safety risk.
Control	Load restraint verification before dispatch
Evidence	Load photographs, checklist, load type, restraint method, supervisor approval
Data sources	Driver app, WMS, TMS, Incident system
Required data fields	load_id, trip_id, site_id, load_type, restraint_method, photo_id, approval_status, incident_id
Test type	AI
Test logic	AI/rule identifies mismatch between load type and restraint method.
PSOE	Operating/Effective
Severity	High
Finding	Potential load-restraint control failure
Remediation	Re-secure load; obtain approval; retrain site/driver where required; monitor recurrence

DRV-001 - Licence expired

Field	Value
Domain	Driver Competency
Obligation	Ensure drivers are competent, authorised and appropriately trained for assigned work.
Risk	Unqualified or untrained driver operates vehicle or performs safety-critical task.
Control	Licence, induction and competency verification control
Evidence	Driver master, licence, training, induction and assignment record
Data sources	HR, LMS, Driver app, TMS, Supplier portal
Required data fields	driver_id, licence_class, licence_expiry, vehicle_class, training_id, training_expiry,

	induction_status, supplier_id
Test type	Deterministic
Test logic	Driver assigned with expired licence.
PSOE	Present/Operating
Severity	Critical
Finding	Potential driver competency control failure
Remediation	Block assignment where required; update records; complete training/induction; verify before release

CAR-005 - Repeat incident despite closed CAR

Field	Value
Domain	Incidents & Corrective Actions
Obligation	Identify, investigate, remediate and verify transport safety issues and control failures.
Risk	Incidents recur or remain unresolved due to weak corrective action.
Control	Incident, corrective action and effectiveness monitoring control
Evidence	Incident reports, investigations, CARs, closure evidence, escalation records
Data sources	Incident system, GRC, Email/DMS, TMS, Maintenance
Required data fields	incident_id, severity, root_cause, car_id, owner, due_date, closure_evidence_id, recurrence_tag, escalation_record
Test type	Analytical
Test logic	Same/root-related incident recurs after remediation.
PSOE	Operating/Effective
Severity	High
Finding	Potential incident/CAR control failure
Remediation	Assign/complete investigation; update CAR;

capture closure evidence; verify effectiveness and recurrence

SUP-001 - Supplier onboarding evidence missing

Field	Value
Domain	Supplier / CoR Assurance
Obligation	Manage CoR and assurance risks across outsourced transport arrangements.
Risk	Customer or contracting party lacks visibility of supplier controls and evidence.
Control	Supplier onboarding, passport and assurance monitoring control
Evidence	Supplier profile, contracts, certificates, attestations, shared assurance status, findings
Data sources	Supplier portal, Procurement/ERP, Contract system, DMS, Assurance platform
Required data fields	supplier_id, contract_id, insurance_expiry, certificate_status, assurance_status, passport_last_updated, subcontractor_id, open_findings
Test type	Deterministic
Test logic	Supplier is active without required assurance evidence.
PSOE	Present/Suitable/Operating
Severity	High
Finding	Potential supplier assurance control failure
Remediation	Request evidence; restrict work allocation if needed; update passport; escalate unresolved supplier risk

9. User journeys

Compliance Manager - fatigue finding: Opens Findings Inbox -> filters High severity -> reviews FAT-001 evidence -> confirms finding -> assigns CAR -> monitors next 30 days for recurrence.

Fleet Manager - maintenance exception: Receives MNT-002 alert -> checks vehicle, defect and dispatch history -> grounds vehicle -> uploads repair evidence -> releases after inspection.

Auditor - workpaper review: Opens Audit Readiness -> selects Fatigue domain -> sees applicable controls, evidence population, exceptions and samples -> requests extra evidence or records conclusion.

Enterprise shipper - supplier assurance: Opens supplier dashboard -> views Compliance Passports -> prioritises amber/red suppliers -> requests evidence or remediation from carrier.

10. Integration architecture

Integration	Core data	Priority	Purpose
TMS	Jobs, trips, routes, schedules, customers, loads	P0	Scheduling, journey and customer risk.
EWD / work diary	Work/rest, exceptions, approvals	P0	Fatigue assurance.
Telematics / GPS	Location, speed, route, events	P0	Speed, route and schedule correlation.
Fleet maintenance	Services, defects, repairs, inspections	P0	Roadworthiness assurance.
HR / LMS	Drivers, licences, training, induction	P1	Competency assurance.
WMS / loading	Load, site, dispatch, photos, timing	P1	Mass/loading and load restraint.
Weighbridge / permit register	Mass, axle, permits, conditions	P1	Mass/dimension assurance.
Supplier / procurement	Suppliers, contracts, certificates, insurance	P2	Compliance Passport and supplier assurance.
GRC / DMS	Policies, procedures, incidents, CARs	P1	Document, control and remediation evidence.

11. AI architecture and guardrails

- Deterministic rules handle known configured thresholds, expiries, missing fields and status checks.
- Analytics detect trends, clusters, recurrence and correlations such as speed events after loading delays.
- AI supports document classification, evidence mapping, root-cause hypotheses, corrective-action adequacy checks, audit narratives and grounded Q&A.
- AI outputs are stored with model, prompt, source evidence, confidence and human review trail.

- High-risk findings and audit conclusions require human review and cannot be auto-closed by AI.

12. Security, privacy and trust requirements

- Tenant isolation, RBAC/ABAC, MFA and SSO/SAML/OIDC.
- Least-privilege connector scopes and customer-controlled evidence sharing.
- Immutable audit logs for evidence collection, tests, AI outputs, user reviews and closure decisions.
- Encryption in transit and at rest, Australian hosting option and configurable retention.
- Product operating model should be designed for SOC 2 / ISO 27001 readiness even if certification comes later.

13. Seven MVP screens

Assurance Overview: PSOE by control/domain, high-risk findings, overdue CARs and evidence freshness.

Control Explorer: Regulation -> obligation -> risk -> control -> evidence -> tests -> findings.

Findings Inbox: Potential/confirmed findings, severity, status, reviewer and evidence bundle.

Evidence Explorer: Source evidence with lineage, source ID, timestamp, entity linkage and test result.

Corrective Actions: Owner, due date, remediation, closure evidence and effectiveness monitoring.

Audit Readiness: Workpaper pack by control with population, samples, exceptions, CARs and comments.

Ask Compliance: Grounded Q&A over controls, evidence, findings and suppliers.

14. Product phases

Phase 0 - Foundation: Control graph, applicability engine, evidence schema, security baseline, connector framework, design-partner validation.

Phase 1 - Operator MVP: 8 domains, 48 tests, P0 integrations, findings, CARs, PSOE dashboard, audit pack and basic copilot.

Phase 2 - Audit Workspace: Auditor roles, evidence requests, sample support, workpaper comments, findings sign-off and export.

Phase 3 - Supplier Assurance: Compliance Passport, supplier onboarding, evidence sharing, supplier dashboard and customer-specific controls.

Phase 4 - Expanded Compliance: WHS, dangerous goods, environmental/customer standards and WA/NT rule packs as separate modules.

Phase 5 - Compliance Network: Multi-party assurance relationships across enterprise shippers, 3PLs, carriers and subcontractors.

15. MVP acceptance criteria

- Can ingest sample TMS/EWD/GPS/fleet data and map it to driver, vehicle, trip and supplier entities.
- Can run at least 25 deterministic tests and 10 analytical tests with versioned results.
- Can generate potential findings with source evidence and human review workflow.
- Can create CARs, require closure evidence and monitor recurrence.
- Can produce an audit-ready workpaper pack for five or more control domains.
- Can answer natural-language questions only when grounded in permitted evidence.
- Can show PSOE status by control and evidence freshness.
- Can maintain immutable audit trail for evidence, tests, reviews and remediation.

16. Sources and benchmark inputs

- NHVR - Heavy Vehicle Accreditation scheme: <https://www.nhvr.gov.au/safety-accreditation-compliance/heavy-vehicle-accreditation-scheme>
- NHVR - HVA scheme audits and PSOE: <https://www.nhvr.gov.au/safety-accreditation-compliance/heavy-vehicle-accreditation-scheme/about-the-hva-scheme/hva-scheme-audits>
- NHVR - Chain of Responsibility: <https://www.nhvr.gov.au/safety-accreditation-compliance/chain-of-responsibility>
- NHVR - Parties in the Chain of Responsibility: <https://www.nhvr.gov.au/safety-accreditation-compliance/chain-of-responsibility/the-primary-duty/parties-in-the-cor>
- NHVR - HVNL and regulations: <https://www.nhvr.gov.au/law-policies/heavy-vehicle-national-law-and-regulations>
- NHVR - Fatigue management: <https://www.nhvr.gov.au/safety-accreditation-compliance/fatigue-management>
- NHVR - Electronic Work Diary: <https://www.nhvr.gov.au/safety-accreditation-compliance/fatigue-management/electronic-work-diary>
- NHVR - Mass, Dimension and Loading: <https://www.nhvr.gov.au/road-access/mass-dimension-and-loading>
- NHVR - Loading requirements: <https://www.nhvr.gov.au/road-access/loading>
- ABS - Counts of Australian Businesses, 2024-25: <https://www.abs.gov.au/statistics/economy/business-indicators/counts-australian-businesses-including-entries-and-exits/latest-release>
- ABS - Australian Industry, 2024-25: <https://www.abs.gov.au/statistics/industry/industry-overview/australian-industry/latest-release>
- Modern trust/compliance automation benchmark - automated evidence, continuous monitoring, audit readiness: <https://www.vanta.com/products/automated-compliance>
- Modern trust/compliance automation benchmark - automated evidence, control monitoring, remediation: <https://drata.com/products/compliance-automation>